

Coin Rise in Viscous Fluid

Measuring Terminal Velocity and Fluid Viscosity

Name(s): _____

Learning Objectives

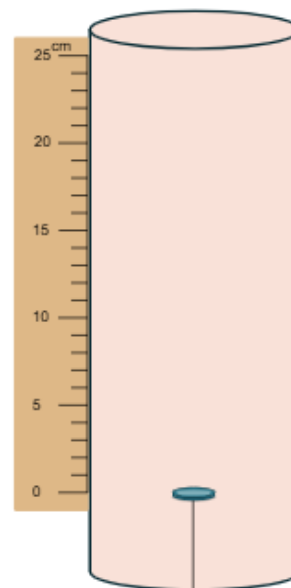
1. Define terminal velocity and describe how to measure it experimentally.
2. Compare the terminal velocity of a disk falling face-down versus edge-on.
3. Use the measured terminal velocity to estimate the viscosity of the fluid.

Experimental Setup

A 3D-printed polystyrene quarter is released into a column of viscous corn syrup, either **face-down** (disk face perpendicular to gravity) or **edge-on** (disk face parallel to gravity). Using the release button, start a stopwatch and record the distance traveled from the release point versus elapsed time. From these data, estimate the terminal velocity for each orientation and use it to infer the fluid viscosity.

Table 1. Key parameters used in the analysis

Symbol	Description	Value	Unit
a	Radius of coin (look up)		cm
t	Thickness of coin (look up)		cm
ρ_e	Density of coin (look up)		g cm^{-3}
ρ	Density of corn syrup	1.33	g cm^{-3}
μ	Viscosity of fluid (to find)	—	cP
g	Gravitational constant	980	cm s^{-2}



Before the Experiment

1. Which orientation do you predict will reach a higher terminal velocity: face-down or edge-on? Explain your reasoning in terms of drag forces.

2. Set the coin orientation to edge-down, release the coin (green Release button) and start a stopwatch. Measure the elapsed time and distance for 6 or 7 distances. Record the values in Table 3.

Table 3. Edge-On Orientation

Elapsed Time (s)	Distance (cm)

After the Experiment

1. **Velocity from data.** Plot distance versus time for each orientation (or compute successive slopes $\Delta d/\Delta t$). At what time does each curve become approximately linear? Estimate the terminal velocity U_{fd} (face-down) and U_{eo} (edge-on) from the late-time slope.

Face-down: $U_{fd} \approx$ _____ cm s^{-1} Edge-on: $U_{eo} \approx$ _____ cm s^{-1}

Is one faster than the other? Does this match your prediction from Question 1?

2. **Force balance and theoretical terminal velocities.** At terminal velocity, the net buoyant force equals the drag. The Stokes-flow drag on a thin disk of radius a is:

$$F_{fd} = 16\mu aU \quad F_{eo} = (32/3)\mu aU$$

Perform a force balance to derive expressions for U_{fd} and U_{eo} in terms of ρ_e , ρ , a , t , μ , and g . Show your work.

What is the predicted ratio U_{fd}/U_{eo} ? Does your measured ratio agree?

3. **Coin dimensions.** Look up the radius a and thickness t of a US quarter and density ρ_e of polystyrene. Record the values in Table 1.

4. **Viscosity.** Use your measured terminal velocity and the expression from Question 2 to solve for μ . Compute μ from *both* orientations.

From face-down: $\mu \approx$ _____ cP From edge-on: $\mu \approx$ _____ cP

5. **Reynolds number check.** Calculate the Reynolds number $Re = \rho a U / \mu$ for each orientation. Is $Re \ll 1$? Does this justify the Stokes drag formulae in Question 2?